Corrections and Clarifications
FFGFT / T0 Time-Mass Duality

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Contents

Preliminary Note

Status: C1–C6, R1–R40 (R16/R17 partial, R20/R30/R32 open, R29 partial, R31 partially derived, R33 declared, R34 clarified; R35/R36 added; R37 revised; C4 (factor 3) added, C5/R38 after external audit; R39 H_0/R_H status clarified; C6 redshift achromatic; R40 absolute values carry correction via bridge constants (v/E_0/factors), ratios exact; 100 (RG amplification) $\neq$ 3609 (φ-scale recursion, open); coupling backdoor) — as of 15 June 2026.

The corrections (C1–C3) and refinements (R1–R13) listed in this document have been incorporated into the respective source documents. The last open item — C2 (tau prefactor r_τ) — was fully completed on 27 May 2026: besides the items reported on 26 May in Doc. 016, Doc. 046 and Doc. 116, this also affected remaining *derived* values in Doc. 046 DE (energy $18{,}8 \to 18{,}0$ meV) as well as the entire **English** parallel version of Docs. 016, 046 and 116, which had not yet been updated on 26 May. For details and the target state see the section "Status of incorporation" under C2.

This document is retained as a historical archive. For r_τ the value $25/9$ recorded here is authoritative.

Addendum (2 June 2026): Added and incorporated is **R14a** (formulation of the redshift in Doc. 041): implemented in Doc. 041 (DE+EN).

Addendum (6 June 2026): Carried over from the CMB work (Doc. 267 Cosmological Degeneracy, Doc. 268 CMB peaks from T^4): **R18** (ΛCDM/FFGFT as two readings, theoretical not empirical), **R20** (absolute cosmological scale: *form* fixed as the ξ ratio R_H/ℓ_P, the *factor* $41/4$ a declared external calibration from ΛCDM, not derived, open), **R29** (peak selection rule, only *partially* answered – the factor-3 filter does *not* exclude $|n|^2 = 30$), **R30** (rigorous C_ℓ projection; the naive spherical Bessel sum is *insufficient*, open) and **R31** (geometric anharmonicity from the codimension-1 projection – leading behaviour derived forward, small residual open). New **R32** (statements presenting H_0 as FFGFT-derived are to be cleaned up corpus-wide – noted for now). The previously separate items *R19* (CMB path lengthening) and *R28* (exponent for z_*) are merged into **R14a** resp. **R20** and do not appear as their own rows.

Addendum (8 June 2026): New **R33** – the magnitude-carrying factor 5^4 in $\xi = 1/(3\cdot 2^2\cdot 5^4)$ is *not* derived forward but grounded intuitively (harmonic / Euler Tonnetz) resp. empirically (Higgs sector); the *form* ($4/3$, 3, 2^2) is forward, the *magnitude* (5^4) remains at the intuitive derivation and is openly *declared* as such. Structural parallel to R20 (form fixed, factor open).

Addendum (8 June 2026, II): New **R34** (dimensional structure). Doc. 181 presents $D = 4$ as "following necessarily"; clarified: **3+1 is empirical input** (no theory derives the dimension count). Forward are only (i) periodicity, (ii) torus-not-sphere ($\pi_1 \neq 0$), (iii) minimal sufficiency (4 instead of 5 via the duality $\tilde{T}m = 1$), (iv) the lossless Hilbert representation (Type III). At the same time recorded: the SI conversions are *indispensable* for the exponent structure (*power of* c, power of time) – c is the space-time bridge (T^3 ↔time), $\hbar$ the mass-time bridge (S^1_m); $c = \hbar = 1$ collapses L, T, M and destroys the powers. The four-circle structure *is* this bookkeeping. Signature as R20/R33 (structure forward, number input).

Addendum (10 June 2026): New **R37** (use of φ in various documents – clarification of the levels). Occasion: Doc. 275 shows that $1/\varphi$ is the fixed point of the ξ damping recursion after $k^* = \log(\varphi)/\xi \approx 3609$ scale levels. In the process it turned out that φ appears in the corpus in *three different roles* that must not be conflated. R37 documents this separation and augments the ξ^N/φ^N triviality warning (R35) by the role level. Note: R35 (ξ^N triviality) and R36 (R_H interpretation) had been declared via the changelog (Updates 16/17); their table entries were added on 10 June 2026.

Addendum (11 June 2026): New **C5** (three implementation errors in the validation script v3 for Doc. 275: point duplicates, seed bug, detector bias – all fixed in v4, affected results retracted) and **R38** (k^* result downgraded from "fixed point without a free parameter" to "pass-through point, R35 applies"; R37 level (2) revised accordingly). Both entries go back to the independent audit by P. Stoychev (IPI list, 11 June 2026) – the first external audit entry of the register.

Addendum (14 June 2026): New **R39** (status of H_0/R_H clarified). Occasion: Doc. 277. It is clarified that the relation Planck length↔ R_H (exponent $41/4$) follows from *no* framework out of first principles; ΛCDM fixes it via the measured H_0 (itself circular, Doc. 267), FFGFT adopts the established value for comparability. Hence the cosmic scale is a *shared empirical reference point*, not an FFGFT-specific gap and not an asymmetric disadvantage relative to ΛCDM (R20/R32/R36 remain valid but are no longer read as a weakness). The only remaining open point is the forward derivation of $41/4$ from ξ – symmetric, since no framework has it; the only model asymmetry is the dark sector (carried by ΛCDM alone). At the same time a numbering conflict was resolved: the v3 script audit (initially recorded as C4 in the addendum of 11 June) is now **C5**; **C4** denotes the older factor-3 correction (Doc. 267/268, 3D observer instead of fractal dispersion), added as a table row.

Addendum (15 June 2026): New **C6** (redshift is *achromatic*, not wavelength-dependent). Occasion: Doc. 280 and the finite-element simulation of the fractal path-lengthening. It corrects the earlier chromatic presentation ($z(\lambda) \propto \lambda$, "UV stronger than radio") in Doc. 004, 025, 030, 039, 041, 053, 061, 063, 068, 081. The mature mechanism (geometric path-lengthening) stretches all wavelengths by the same factor; the observed cosmological redshift is achromatic, which FFGFT gets right. The earlier "wavelength-dependent discriminator" drops out – it moves to the "fits" column (data degeneracy, Doc. 267).

Addendum (15 June 2026): New **R40** (absolute values carry the fractal/SI correction via the bridge constants; ratios are exact). In FFGFT only the *dimensionless ratios* are exact. Absolute values carry the fractal/SI correction because it is absorbed into the dimensionful *bridge constants*: v (masses $m = r\xi^p v$), $E_0 = 7{,}398$ MeV ($\alpha = \xi E_0^2$) and the factors $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; canonical script `calc_De.py`, which carries *no* explicit K). In ratios v/E_0/factors cancel ($m_\mu/m_e = 206{,}768$, correction-free). Factored out, this correction is $K_{\text{frak}} = 1 - 100\,\xi \approx 0{,}987$ (Doc. 011/012; in Doc. 133 derived from D_f-consistency and an RG run over 17 orders of magnitude – *not an adjusted correction*). **Do NOT confuse two numbers:** the *100* in K_{frak} is the RG amplification of the $\sim 1{,}3\,\%$ correction (Doc. 133); the *3609* (Doc. 275) is the depth of the ξ-scale recursion $r(k) = \frac{D_f}{3}(1-\xi)^k$ at which the sequence *passes* $1/\varphi$. There $1/\varphi$ is a *passage point, not a fixed point* (the only fixed point is 0); since the sequence passes *every* value, 3609 is unanchored unless $1/\varphi$ is independently motivated (status open, R37). Moreover further mutual couplings are not excluded (⇒ even ratios are leading order; *backdoor*): a deviation from the simple rational can always be attributed to an unmodelled coupling – ratio tests confirm (Koide 10^{-5} at $2/3$) but barely falsify. Clean falsification only via a *forward*-fixed new prediction ($41/4 \to H_0 \approx 66{,}2$ against the H_0 tension).

Addendum (2 July 2026): New **P41** (redshift drift Sandage–Loeb checked and not booked: amplitude test not degeneracy-proof at the shared scale $c/R_H = H_0$, only a conditional shape test) and **P42** (lepton sector: one declared reference point m_μ/m_e; $2/9$ as the rational address of θ_{ref}, sharpened prediction $m_\tau = 1776.9690$ MeV; occasion: precision analysis Doc. 292). At the same time the doubly assigned heading "Refinement 14" was resolved into **R14a** (redshift Doc. 041, incl. table row) and **R14b** (CMB anharmonicity, covering R29–R31).

This document lists corrections and refinements to already published documents of the FFGFT series. Older documents are *not* changed. Where a contradiction exists between an older and a newer document, the version recorded here is authoritative.

Each entry contains: the affected document and location, the erroneous expression, the correct expression, and a short justification.

.1 Correction 1: Prefactor in the Higgs derivation of ξ

Affected document

Doc. 049 (T0 Lagrangian and Higgs integration), HTML interim version and older working versions.

Erroneous expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1{,}0 \times 10^{-5} \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1{,}33 \times 10^{-4}} \tag{190.1}$$

with the experimental inputs:

$$\begin{aligned} m_h &= 125{,}25\,\text{GeV} \quad \text{(Higgs mass)},\\ v &= 246{,}22\,\text{GeV} \quad \text{(vacuum expectation value)},\\ \lambda_h &= \frac{m_h^2}{2v^2} \approx 0{,}129 \quad \text{(Higgs self-coupling)}. \end{aligned}$$

Justification

The prefactor $64\pi^4$ stems from a typo in the early HTML visualization. It yields $\xi \approx 10^{-5}$, which deviates by an order of magnitude from the geometrically derived value $\xi = 4/3 \cdot 10^{-4}$ and does not reproduce the natural constants. The correct prefactor $16\pi^3$ gives $\xi \approx 1{,}33 \times 10^{-4}$, consistent with the independently determined value from the proton-electron mass ratio (Doc. 009).

.2 Correction 2: Tau Yukawa prefactor r_τ

Affected documents

The prefactor r_τ appears in several documents. Complete list:

- Doc. 116 (Koide formula as a consequence of T0 theory), lepton-parameter table.
- Doc. 016 (Complete calculations), mass table and r-parameter list.
- Doc. 046 (Particle masses), tau-neutrino row (double-ξ construction containing r_τ).

Already correctly updated at the time of the first correction were Docs. 006, 028, 158, 210 as well as the theorem and proof part of Doc. 116.

Erroneous expression

$$r_\tau = \frac{8}{3} \approx 2{,}667 \quad \Rightarrow \quad m_\tau \approx 1712\,\text{MeV} \quad (-3{,}6\,\%\text{ dev.}) \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2{,}778} \quad \Rightarrow \quad m_\tau \approx 1783\,\text{MeV} \quad (+0{,}4\,\%\text{ dev.}) \tag{190.2}$$

The lepton masses follow generally from:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \tag{190.3}$$

with the exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ (step size $\Delta p = 1/3$) and the vacuum expectation value v.

Justification

$r_\tau = 8/3$ was a remnant from an earlier iteration step. Doc. 158 (Koide-to-g-2 bridge) already uses $r_\tau = 25/9$, which matches the tau mass numerically. Moreover $25/9 = (5/3)^2$ has a geometric interpretation as the square of the pure sixth in the natural harmonic series, consistent with the musical resonance analogy (Doc. 060).

Complete corrected prefactor table

Lepton	Exponent p	Prefactor r	m [MeV]
Electron	$3/2$	$4/3$	0,511
Muon	1	$16/5$	105,7
Tau	$2/3$	$25/9$	1783

Status of incorporation (Addendum 26 May 2026)

A renewed corpus-wide search found **remaining** $8/3$ **residues**. The correction had thus *not been carried through cleanly*; the original entry named only Doc. 116 as affected and overlooked Docs. 016 and 046.

Document / location	Finding	Target
Doc. 116, lepton-parameter tab. (contradicts its own theorem)	$r_\tau = 8/3$	$25/9$
Doc. 016, mass tab. and r-list ($m_\tau = 1712{,}1$, $-3{,}64\,\%$)	$r_\tau = 8/3$	$25/9$ (≈ 1783, $+0{,}3\,\%$)
Doc. 046, tau-neutrino (3 places) (param. row, double-ξ calc., second ν table; cf. Doc. 006)	$r = 8/3$	$25/9$

Carried out on 26 May 2026: All of the above places were brought to $25/9$; in Doc. 016 the derived tau mass ($1712{,}1 \to 1783{,}4$ MeV) and the deviation ($3{,}64 \to 0{,}37\,\%$) were recomputed, in Doc. 046 the ξ_{ν_τ} calculation ($128/27 \to 400/81$, $E = 18{,}8 \to 18{,}0$ meV). C2 is thereby fully incorporated.

Addendum 27 May 2026 (actual extent): An independent full check on 27 May found that the incorporation on 26 May was *not* complete. Two points were still open:

- **Doc. 046 DE** still carried, in the *derived* tables (compatibility, method-equivalence and falsification tables), the old energy value $18{,}8$ meV as well as the decimal form $r_\tau = 2{,}768$. The main calculation ($\xi_{\nu_\tau} = 400/81$) was already correct; only the downstream values had not been updated. Corrected to $18{,}0$ meV resp. $2{,}778$.

- The **English parallel version** of Docs. 016, 046 and 116 had *not* been updated at all on 26 May (DE and EN must be consistent). Corrected: 016 EN (mass table $8/3 \to 25/9$, $1712{,}1 \to 1783{,}4$, $3{,}64 \to 0{,}37\,\%$; r-list), 046 EN (both ν_τ rows, ξ_{ν_τ} calc. $128/27 \to 400/81$, three E values, method equivalence), 116 EN (tau row $8/3 \to 25/9$).

Corpus-wide residual search afterwards: **0 remaining** $8/3$ **places** in 016/046/116 (DE+EN). All four files recompiled (Kindle 6×9). C2 has been *fully* incorporated since 27 May 2026 — DE and EN.

.3 Correction 3: Base formula for g-2 of the electron

Affected document

T0-Explorer (HTML visualization), first version; older interim versions of Doc. 018.

Erroneous expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}} \qquad \text{(old – wrong)}$$

Correct expression

$$\boxed{a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}}} \tag{190.4}$$

with $f = 1/\xi \approx 7500$, $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2{,}224$ and $S_3 = 2\pi^2 \approx 19{,}74$ (surface of the 4D unit sphere, i.e. of the 3D boundary of the torus).

Justification

The earlier prefactor 4π corresponds to the surface of the 2D sphere in 3D space, not to the geometrically correct 3D surface of the 4D torus. $S_3 = 2\pi^2$ is the correct value from the torus geometry. The fractal corrections for muon and tau continue to use 4π:

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}}, \tag{190.5}$$

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}. \tag{190.6}$$

.4 Refinement 1: Accuracy of the Koide formula

Affected documents

Doc. 116 (abstract, l. 14): $\Delta Q < 0{,}00003\,\%$
Doc. 158 (abstract): “experimentally confirmed to $0{,}001\,\%$”

Correct placement

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.7}$$

is experimentally confirmed to better than $0{,}001\,\%$ (PDG values 2022). The figure $< 0{,}00003\,\%$ in Doc. 116 refers to the internal consistency of the T0 formulas, not to the experimental measurement uncertainty. Both statements are correct in content but are different quantities. In external communication $0{,}001\,\%$ is the appropriate figure.

Note

The T0 individual masses agree with experiment to $\sim 1\,\%$; the high precision of $Q = 2/3$ arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

.5 Refinement 2: $\hbar$ from the Higgs field – derivation chain

Context

The Planck constant $\hbar$ is derived differently in various documents. Here the complete, consistent derivation chain is recorded.

Authoritative derivation chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \tag{190.8}$$

The time-mass binding condition provides the physical justification:

$$T(x,t) \cdot m(x,t) = 1 \quad \Longrightarrow \quad T = \frac{\hbar}{m\,c^2} \quad \Longrightarrow \quad E = \hbar\omega. \tag{190.9}$$

$h = 2\pi\hbar$ is thereby *not* a free constant but a geometric consequence of ξ and ℓ_P. The numerical SI value $h \approx 6{,}626 \times 10^{-34}$ J·s is the calibration result of this relation onto the SI base units.

.6 Refinement 3: Two ξ parameters – two different spaces

Context

In Docs. 173–176 (quantum dots, spin qubits, qubit state spaces, Shor algorithm) two numerically different ξ values appear. Earlier documents did not keep them consistently separate, which can cause confusion.

Authoritative distinction

Parameter	Value	Domain	Meaning
ξ_0 (geometric)	$\frac{4}{30000} \approx 1{,}33 \times 10^{-4}$	3D geometric space	length hierarchy of the torus; holds in physical position space
ξ_{Higgs} (algebraic)	$\approx 1{,}04 \times 10^{-5}$	number / quantum state space	algebraic field corrections; Higgs sector; decoherence rates

Justification

$\xi_0 = 4/30000$ follows from the torus geometry of physical space (geometric derivation, independent of SI measurements). ξ_{Higgs} follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations, T_2 hierarchy).

The two ξ values are not a contradiction, but describe complementary aspects: geometry of space (ξ_0) vs. algebra of the state space (ξ_{Higgs}). In external communication it must always be made clear which parameter is meant.

.7 Refinement 4: L_0 and the Schwarzschild interpretation

Affected documents

Older versions of Docs. 180–182 (Lagrangian derivation, torus geometry, maximal universe scale).

Erroneous approach

Earlier versions interpreted $L_0 = \xi_0 \cdot \ell_P$ as the Schwarzschild radius of a fundamental minimal entity. This interpretation was explicitly discarded in Doc. 182 (2026 revision).

Correct placement

L_0 is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 = \xi_0 \cdot \ell_P \approx 2{,}15 \times 10^{-39}\,\text{m}. \tag{190.10}$$

It defines the lowest scale level of the ξ hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius R_H is the system-specific upper bound of the cosmological sector – not a universal maximal size. Every scale has its own system-specific upper bound.

Remark: Schwarzschild as a consistency check

The Schwarzschild connection remains valid as an **internal consistency check**: an object with mass $M_0 = \xi_0 \cdot m_P/2$ would have a Schwarzschild radius of exactly L_0 – without a free parameter. If M_0 matches a scale level of the ξ hierarchy, that is a nontrivial self-consistency test. The connection is *a probe, not a derivation*.

.8 Refinement 5: Claim and scope of the mass formulas

Context

Several FFGFT documents describe mass derivations for leptons and other particles. The claim of these derivations must be communicated clearly.

Authoritative formulation

The FFGFT mass formulas (e.g. $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$) yield values that agree with experiment to $\sim 1\,\%$. This is **not a precision calculation** but a structural proof:

- The mass formulas *follow geometrically* from ξ and v – they are not fitted.
- The remaining $\sim 1\,\%$ deviations reflect measurement uncertainties and approximations, not errors of the structure.
- Numerically better calculations than the Standard Model are *not* claimed.

In external texts the following is to be avoided: "FFGFT computes the masses exactly." Correct is: "FFGFT *derives* the mass structure from a single parameter ξ and reproduces the orders of magnitude to $\sim 1\,\%$."

.9 Refinement 6: Koide formula only with original values

Authoritative rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.11}$$

may only be evaluated with experimentally measured masses (PDG values).

Not admissible: inserting the symbolic FFGFT terms $r_\ell \cdot \xi^{p_\ell} \cdot v$ directly into the formula. The $\sqrt{m_i}$ operation generates cross terms with mixed ξ powers that are foreign to the torus structure space. Any such insertion yields $Q \approx 0{,}6677 \neq 2/3$ and is not a contradiction to the theory but a category error.

Admissible: evaluate the FFGFT terms numerically (e.g. $m_\tau = 1783$ MeV) and insert this result into the Koide formula like a measured value.

Mathematical background: The r_ℓ terms are geometric invariants of their torus modes. The set of admissible pairs (r_i, p_i) is not closed under multiplication: $r_e \cdot r_\mu = 64/15$ belongs to no torus mode. This is the non-closure theorem (Doc. 193). The general analysis of the composition limits is found in Doc. 192.

.10 Refinement 7: Terminology "renormalization group" and scale structure

Affected documents

- Doc. 005 (T0-tm extension): "RG flow", "renormalization-group factor", "renormalization-group invariance", subsection "renormalization-group treatment".
- Doc. 008 (ξ and e): subsection "modified renormalization-group equations".
- Doc. 019 (T0 Lagrangian, older version): subsection "renormalization-group equations".
- Doc. 086 (document overview): "renormalization group as a flow in parameter space".

- Doc. 189 (torus derivation): “cumulative renormalization-group running”.
- Doc. 202 (FFGFT field theory, overall), §18: section title “The renormalization group”.

Previous formulation

In the documents named, the ξ_0-modified scale running of the coupling

$$\frac{d\alpha}{d\ln\mu} = \beta_0\,\alpha^2\Big(1+\xi_0\,\ln\frac{\mu}{E_0}\Big) \qquad \text{(old – imprecise)}$$

and the associated scale dependence are called the *renormalization group*, at times in immediate proximity to statements that characterize ξ_0 explicitly as a topologically fixed geometric constant.

Refined formulation

In FFGFT the scale dependence of the coupling is *not* the result of a renormalization procedure but a direct consequence of the recursion $\mathscr{R}$ (Doc. 203) on the fractal torus geometry. The correct designation is therefore:

$$\boxed{\begin{array}{c}\text{scale structure from the recursion}\\ \equiv\ \text{recursive } \alpha \text{ running via } \mathscr{R}\end{array}} \qquad (190.12)$$

Concretely:

- The term $\xi_0\,\ln(\mu/E_0)$ arises from the iterated application of the recursion $\mathscr{R}$ to the mode structure, not from a counter-term subtraction.
- ξ_0 is *not* a fixed point of a β function but a topologically fixed constant ($\xi_0 = 4/30000$, Doc. 180).
- Stability, scale flow and UV finiteness are three aspects of the *same* recursive structure – not separate mechanisms.

Language convention

Avoid	Use
renormalization group (implies a QFT procedure)	scale structure from the recursion
RG flow / RG running	recursive α running; $\mathscr{R}$-induced scale correction
renormalization-group equations	scale equations from $\mathscr{R}$
renormalization-group invariance	recursive scale invariance
renormalization-group fixed point	fixed point of the recursion: $\mathscr{R}(\Phi_*) = \Phi_*$

Justification

The designation “renormalization group” imports a tool of standard quantum field theory that rests on a completely different principle: there, UV divergences are absorbed by a procedure, and the parameters *run* under a formal scale transformation. In FFGFT neither the divergence problem (cf. Doc. 159, “without artificial renormalization”) nor a free scale parameter to be absorbed exists. The scale dependence is a *structural property* of the recursion, not a subsequent adjustment step.

The FFGFT-native view is already correctly formulated in Doc. 159 ("renormalization as a symptom, not a solution") and Doc. 189 ("geometrically determined, not by renormalization"); this refinement serves the uniform terminology across the older Lagrangian and ξ-derivation series (Docs. 005, 008, 019) as well as across Doc. 202.

The underlying mechanism is worked out in Doc. 203 (recursion operator $\mathscr{R}$) and Doc. 204 (fractal boundary condition).

.11 Refinement 8: Terminology "fractal renormalization"

Affected documents

Docs. 005, 008, 012, 014 (with 19 occurrences the most frequent source), 041, 060, 133, 141, 159, 181.

Previous formulation

"Fractal renormalization" is used as an FFGFT-internal term for the geometric correction factors in the conversion between natural and SI units as well as for scale-dependent corrections on the fractal torus.

Refined formulation

Since the word "renormalization", even in compound form, carries the QFT connotation of a subtraction or adjustment procedure, the term is replaced by context-specific alternatives:

Context	**Authoritative designation**
unit conversion natural ↔ SI (esp. Doc. 014)	geometric scale adjustment
scale-dependent corrections on the fractal torus (Docs. 133, 159, 181)	fractal correction (matches the title of Doc. 133)
general scale transformation by recursion	$\mathscr{R}$-induced scale correction

Justification

The internal term "fractal renormalization" was meant to set the procedure clearly apart from QFT renormalization, but it adopts the central word and with it inevitably the connotation of a *correction procedure*. In FFGFT it is not a procedure but the direct numerical manifestation of the fractal geometry. The proposed alternatives avoid this misunderstanding without new terminology – both phrasings are already established in the corpus (Doc. 133 "fractal correction"; "geometric scale adjustment" in the same semantic field as Doc. 014).

.12 Refinement 9: Column "symmetry" and the term "color charge" in the fermion tables

Affected documents

- Doc. 006 (T0 particle masses), "universal table of fermions", column "symmetry".
- Doc. 046 (Particle Masses, English parallel version), same table, column "color".
- Doc. 042 (ξ parameter and particles), table "spatial field patterns", entry "quarks: multi-node bound clusters, confined, color charge".
- Doc. 005 (T0-tm extension), method 2 "extended fractal method with QCD integration": use of $N_c = 3$, α_s, Λ_{QCD} as external inputs.

Previous formulation

In the fermion tables of Doc. 006/046 each row carries a column "symmetry" with the following entries:

Particle	Entry in "symmetry"
electron, muon, tau	lepton number
ν_e, ν_μ, ν_τ	double-ξ
up, charm, strange, top, bottom	color
down	color + isospin

This mixture combines four different concepts in one column: an FFGFT scaling class ("double-ξ"), an SM conserved quantity ("lepton number"), an SM gauge symmetry ("color") and an SM quantum number ("isospin"). In addition, the entries "color charge" in Doc. 042 and $N_c = 3$ in Doc. 005 suggest that the SM color symmetry $SU(3)_C$ is an independent component of FFGFT.

Refined formulation

In FFGFT neither the gauge group $SU(3)_C$ nor a color quantum number exists. Quark masses follow, just like lepton masses, exclusively from the Yukawa resonance formula

$$m_f = r_f \cdot \xi^{p_f} \cdot v \tag{190.13}$$

with r_f a rational prefactor and p_f a rational exponent (step size $\Delta p = 1/3$). This is numerically substantiated by the script `calc-resonanz-leiter.py`: all six quarks hit their PDG masses to $\sim 0{,}3$–$3{,}4\,\%$, *without* use of N_c, α_s, Λ_{QCD} or a color quantum number.

Corrected column structure of the fermion table

The old column "symmetry" is replaced by two clearly separated columns:

Particle	FFGFT scaling class	SM correspondence (informative)
electron, muon, tau	single-ξ	lepton number
ν_e, ν_μ, ν_τ	double-ξ	lepton number, lepton mixing
up, charm, top	cluster-ξ	up-type, $T_3 = +1/2$
down, strange, bottom	cluster-ξ	down-type, $T_3 = -1/2$

Meaning of the classes:

- *single-ξ*: $m = r\xi^p v$ with $p \in \{2/3, 1, 3/2\}$.
- *double-ξ*: $m_\nu = r_\nu \xi^2 m_e$ (additional ξ suppression relative to the charged leptons).
- *cluster-ξ*: $m = r\xi^p v$ with the same exponents $p \in \{-1/3, 1/2, 2/3, 1, 3/2\}$ as for the leptons, but with multi-node binding in the sense of Doc. 049 (cubic self-interaction $\lambda_s(\delta m_q)^3$).

Language convention for quarks

Avoid	Use
"color charge" (implies an $\mathrm{SU}(3)_C$ quantum number)	multi-node binding
"three colors red/green/blue" (SM gauge index)	triple-node cluster (the triplet is geometry, not an $SU(3)$ index)
"$N_c = 3$ colors"	$N_{\text{cluster}} = 3$ (geometric node count in the baryon)
"color confinement"	cluster binding (cubic self-interaction)

Status of Doc. 005, method 2

Doc. 005 contains two calculation methods:

- *Method 1: direct geometric method.* Yukawa form per Eq. (190.13); parameter-free; accuracy $\sim 1\,\%$ for all fermions. **This is the FFGFT method.**
- *Method 2: extended fractal method with QCD integration.* Uses N_c, α_s, Λ_{QCD}, lattice-QCD calibration and ML. **This is a phenomenological bridge construction** for linking to the lattice-QCD literature, not part of the parameter-free FFGFT.

In external communication method 1 is to be cited. Method 2 may be used if explicitly labelled as a "bridge to lattice QCD" or "ML-calibrated extension".

Justification

The designations "color", "color charge", "N_c" import a concept of standard quantum field theory ($SU(3)_C$ as a gauge symmetry, three colors, eight gluons, color confinement as a non-perturbative property). Doc. 049 states explicitly that FFGFT contains none of these elements: "what we call ‚color' becomes high-energy node-binding patterns" (Doc. 049, §3.7), with a single cubic interaction term $\lambda_s(\delta m_q)^3$ instead of eight gluon fields.

The script validation (`calc-resonanz-leiter.py`, `calc_De.py`) shows: all quark masses follow from the same Yukawa formula as the leptons. The column "symmetry" of the fermion table, by contrast, suggests that $SU(3)_C$ is a distinguishing property of the quarks within FFGFT – which is not the case.

The corrected column structure separates the FFGFT-internal (*scaling class*) from the informative SM link (*SM correspondence*), parallel to the separation of "recursive α running" and "renormalization group" in Refinement 7.

.13 Refinement 10: Logical status of $K = 245$ and the factor $1{,}314$ in Doc. 009

Affected document

Doc. 009 (origin of ξ), section "why $K = 245$ is fundamental", subsection "geometric meaning".

Misleading presentation

Doc. 009 lists the following derivation chain for $K = 245$:

- $\varphi^{12} = 321{,}996$
- correction by the fractal structure: $(1-\xi)^2 \approx 0{,}999733$
- result: $321{,}996 \times 0{,}999733 \approx 321{,}87$
- "scaling to the mass range": $321{,}87/1{,}314 \approx 245$

The factor $1{,}314$ appears without derivation or independent physical justification.

Correct logical order

$K = 245$ is *not* derived from φ^{12} and $1{,}314$. The correct logical order is the reverse:

$$K = \frac{R_{pe}}{(4/3)^{\kappa}} = \frac{1836{,}153}{(4/3)^7} \approx 245{,}097 \tag{190.14}$$

K is fixed by the observed proton-electron mass ratio R_{pe} and the exponent $\kappa = 7$. It is an *empirical anchor*, not a result from first principles. The factor $1{,}314$ is defined implicitly as $\varphi^{12}/245 \approx 1{,}314$ and does not constitute an independent derivation of K.

What remains non-trivial

The exponent $\kappa = 7$ is the *only integer* for which the complete e-p-μ triangle closes simultaneously:

κ	$K \times (4/3)^{\kappa}$	agreement with R_{pe}
6	1376,6	$-25\,\%$
7	1835,4	✓ $0{,}04\,\%$
8	2447,2	$+33\,\%$

This uniqueness is a real condition, not a tautology. $K = 245$ then follows from $\kappa = 7$ and the empirical R_{pe}.

Corrected formulation for external communication

- **Avoid:** "$K = 245$ is derived from first principles; no free parameters."
- **Use:** "An empirical anchor: $K = 245$, fixed by R_{pe} and $\kappa = 7$. From this anchor follow five mass ratios with $< 0{,}04\,\%$ accuracy. The φ^{12} observation is a numerological consistency note, not a derivation."

.14 Refinement 11: T_{CMB} correction factor and K_{frac} clarification

Affected documents

Doc. 025 (cosmology), Doc. 003 (foundations), Doc. 133 (fractal correction derivation).

Problem A: K_{frac} does not close the T_{CMB} gap

Doc. 025 derives:

$$T_{\text{CMB}} = \tfrac{16}{9}\xi = \tfrac{4}{16875}\,\text{eV} \approx 2{,}7507\,\text{K} \tag{190.15}$$

The measured value is $T_{\text{CMB}} = 2{,}72548$ K (Planck 2018). The gap is $\approx 0{,}925\,\%$.

$K_{\text{frac}} = 1 - 100\xi = 74/75$ (Doc. 133) does *not* close this gap. Numerically:

$$\tfrac{16}{9}\xi \times K_{\text{frac}} = 2{,}714\,\text{K} \quad (0{,}42\,\%\ \text{below the target value, worse}) \tag{190.16}$$

The required correction factor is $0{,}99083$, achieved by the following formula:

$$\boxed{T_{\text{CMB}} = \tfrac{16}{9}\xi\left(1 - \tfrac{275}{4}\xi\right) = 2{,}725486\,\text{K}} \tag{190.17}$$

Agreement with the measured value: $\Delta = 0{,}0002\,\%$.

The coefficient $275/4 = 68{,}75$ lies close to the tetrahedral number $68 = 72 - 4$ (Doc. 003, section 0.5), but its derivation as a second-order correction is not yet formally established. **This is an open point:** a geometric derivation of $275/4$ is required. The formula above is a numerically verified result that still requires a complete justification.

Problem B: two inconsistent K_{frac} definitions

Doc. 003 gives:

$$K_{\text{frac}}^{(003)} = 1 - \frac{D_{\text{frac}} - 2}{68} = 1 - \frac{0{,}94}{68} \approx 0{,}98530 \tag{190.18}$$

where $D_{\text{frac}} = 2{,}94$ is given without derivation from ξ.

Doc. 133 gives:

$$K_{\text{frac}}^{(133)} = 1 - 100\xi = \tfrac{74}{75} \approx 0{,}98667 \tag{190.19}$$

derived from the cumulative RG running with $D_f = 3 - \xi$.

These values differ by $1{,}37 \times 10^{-3}$ and cannot both be correct. The **primary definition** is Doc. 133: $K_{\text{frac}} = 1 - 100\xi = 74/75$. The value $D_{\text{frac}} = 2{,}94$ in Doc. 003 is an independent approximation not derived from ξ; it uses a different (tetrahedral) model. Doc. 003 is to be understood as a heuristic motivation, not as a defining formula.

Corrected formulation for T_{CMB}

- **Avoid:** "K_{frac} closes the T_{CMB} gap."
- **Use:** "T_{CMB} requires a second-order correction $(1 - 275/4 \cdot \xi)$ that yields $0{,}0002\,\%$ agreement. The geometric origin of $275/4$ is still open."

Addendum: ΔCHSH $\approx 10^{-5}$ in Doc. 230

Doc. 230 claims ΔCHSH $\approx 10^{-5}$ as a geometrically motivated prediction. Doc. 022 gives the formula

$$\mathrm{CHSH}(N) = 2\sqrt{2} \cdot \exp\left(-\xi \frac{\ln N}{D_f}\right) \tag{190.20}$$

where N is the number of measurement runs. This formula yields ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ (not 10^{-5}). ΔCHSH grows monotonically with N; at $N = 2$ (minimum) one already obtains ΔCHSH $\approx 8{,}7 \times 10^{-5}$. The value 10^{-5} from Doc. 230 is not attainable at any finite N under the Doc. 022 formula.

Important distinction (Onur Teker, May 2026): CHSH$(N) \to 0$ (correlation vanishes) requires $N \to \infty$ and is a different condition from ΔCHSH $= 10^{-5}$ (distance from the Tsirelson bound). Both conditions confirm that the Doc. 022 formula cannot reproduce the Doc. 230 value. The prediction ΔCHSH $\approx 10^{-5}$ requires a separate derivation or a suppression mechanism that is not yet documented. **Open point.**

.15 Refinement 12: Two different ΔCHSH observables in Doc. 022 and Doc. 230

Affected documents

Doc. 022 (QFT-ML addendum), Doc. 230 (Hilbert-space translation).

The apparent contradiction

Doc. 022 gives ΔCHSH $\approx 5 \times 10^{-4}$ at $N = 73$ via the formula CHSH$(N) = 2\sqrt{2} \cdot \exp\left(-\xi \ln N/D_f\right)$. Doc. 230 claims ΔCHSH $\approx 10^{-5}$ without derivation. These values differ by a factor of ~ 50.

Possible resolution: cumulative vs. elementary deviation

The two values refer to different observables:

- **Doc. 022** describes the *cumulative* deviation over N measurement runs. It grows monotonically with N: ΔCHSH$(N) \sim \xi \ln(N)/D_f \times 2\sqrt{2}$. This is the experimentally relevant quantity for a Bell test with N runs.
- **Doc. 230** plausibly refers to the *elementary* deviation per single measurement – the phase cost of one θ_4 projection cycle. The natural candidate is:

$$\Delta\mathrm{CHSH}_{\mathrm{elem}} = \frac{\xi}{2\pi} \approx 2{,}1 \times 10^{-5} \approx 10^{-5} \tag{190.21}$$

  These are the Landauer costs per measurement expressed as a CHSH deviation: the θ_4 phase runs through a full 2π cycle per measurement, and each cycle costs ξ in geometric units.

  The ratio between cumulative and elementary deviation at N runs:

$$\frac{\Delta\mathrm{CHSH}(N)}{\Delta\mathrm{CHSH}_{\mathrm{elem}}} \approx N \cdot \frac{2\pi}{D_f} \tag{190.22}$$

At $N = 73$ this gives a factor of ≈ 153, consistent with the observed factor of ~ 50–100 between the two values.

Status

This resolution is a **working hypothesis**, not a proven derivation. The identification $\Delta\text{CHSH}_{\text{elem}} = \xi/(2\pi)$ requires a formal justification from the T⁴ projection geometry. If confirmed, there is no contradiction between Doc. 022 and Doc. 230 – they describe the same physical effect at different integration scales.

Experimental consequence: The cumulative deviation $\Delta\text{CHSH}(N) \approx 5 \times 10^{-4}$ at $N = 73$ is within reach of future loophole-free Bell tests (current resolution $\sim 10^{-2}$; required improvement: factor ~ 20). The elementary deviation $\xi/(2\pi) \approx 2 \times 10^{-5}$ would require single-shot precision beyond current technology.

.16 Refinement 13: Bridge formula between ξ_{FFGFT} and ξ_{UIFT}

Context

In the IPI discussion (May 2026) Onur Teker (UIFT) writes:

> $\xi = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$, derived from three independent measurements (T_{CMB}, k_B, m_e). This is not a hyperparameter – it is a prediction.

This identifies ξ_{UIFT} with $\xi_{\text{FFGFT}} = 4/30000$. Numerically they differ considerably. This refinement documents the exact relation.

The key: FFGFT uses eV as the temperature unit

In FFGFT the natural unit system $\hbar = c = k_B = 1$ holds. Temperatures are expressed in eV (not in Kelvin):

$$T_{\text{CMB}}[\text{eV}] = k_B \cdot T_{\text{CMB}}[\text{K}] = k_B \cdot 2{,}72548\,\text{K} = 2{,}3486 \times 10^{-4}\,\text{eV} \tag{190.23}$$

The FFGFT formula (Doc. 025) then reads – to first order:

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\,\xi_{\text{FFGFT}} = 2{,}3704 \times 10^{-4}\,\text{eV} \qquad (\Delta = 0{,}93\,\%) \tag{190.24}$$

and to second order (Doc. 190 R11):

$$T_{\text{CMB}}[\text{eV}] = \frac{16}{9}\,\xi_{\text{FFGFT}}\left(1 - \frac{275}{4}\,\xi_{\text{FFGFT}}\right) = 2{,}3486 \times 10^{-4}\,\text{eV} \qquad (\Delta = 0{,}0002\,\%) \tag{190.25}$$

Bridge formula

Inserting (190.24) into Onur's expression (first order):

$$\begin{aligned}\xi_{\text{UIFT}} &= \frac{k_B\,T_{\text{CMB}}[\text{K}]\,\ln 2}{m_e c^2} = \frac{T_{\text{CMB}}[\text{eV}] \cdot \ln 2}{m_e c^2/\text{eV}} \\ &= \frac{\frac{16}{9}\,\xi_{\text{FFGFT}} \cdot \ln 2}{m_e c^2/\text{eV}}\end{aligned} \tag{190.26}$$

Rearranged:

$$\boxed{\frac{\xi_{\text{FFGFT}}}{\xi_{\text{UIFT}}} = \frac{m_e c^2/\text{eV}}{\frac{16}{9} \cdot \ln 2} = \frac{510\,999\,\text{eV}}{1{,}2323\,\text{eV}} \approx 414\,684} \tag{190.27}$$

SI conversion table (reconstructed from Doc. 013, archived version)

The following table was documented in older versions of Doc. 013 (SI conversions for the natural FFGFT unit system, since shortened). It is reproduced here because it is essential for the bridge formula.

Quantity	Nat. units	SI / eV
temperature unit $[T]$	$= [E]$	1 eV = $k_B^{-1} \cdot 1$ eV = $11{,}604$ K
T_{CMB} [nat.]	$(16/9) \cdot \xi = 2{,}370 \times 10^{-4}$	$2{,}370 \times 10^{-4}$ eV
T_{CMB} [K] (Planck 2018)	—	$2{,}72548$ K
T_{CMB} [eV] $= k_B \cdot T$[K]	—	$2{,}3486 \times 10^{-4}$ eV
$m_e c^2$	$= m_e$ (nat.)	$510{,}999$ eV $= 0{,}511$ MeV
k_B	$= 1$ (nat.)	$8{,}617 \times 10^{-5}$ eV/K
$\hbar$	$= 1$ (nat.)	$6{,}582 \times 10^{-16}$ eV·s
c	$= 1$ (nat.)	$2{,}998 \times 10^{8}$ m/s

Numerical verification

Quantity	Value	Unit
$\xi_{\text{FFGFT}} = 4/30000$	$1{,}3333 \times 10^{-4}$	dimensionless
$\xi_{\text{UIFT}} = k_B T_{\text{CMB}} \ln 2/(m_e c^2)$	$3{,}1858 \times 10^{-10}$	dimensionless
ratio $\xi_{\text{FFGFT}}/\xi_{\text{UIFT}}$	$418\,521$	—
scale factor $m_e c^2/(\frac{16}{9} \ln 2)$ [eV]	$414\,684$	—
T_{CMB} [K] (Planck 2018)	$2{,}72548$	K
T_{CMB} [eV] $= k_B \cdot T_{\text{CMB}}$[K]	$2{,}3486 \times 10^{-4}$	eV
$(16/9) \cdot \xi$ [eV] (FFGFT 1st order)	$2{,}3704 \times 10^{-4}$	eV
$(16/9) \cdot \xi(1 - 275/4 \cdot \xi)$ [eV] (R11)	$2{,}3486 \times 10^{-4}$	eV
$m_e c^2$	$510\,999$	eV

The residual deviation of $0{,}93\,\%$ between the ratio ($418\,521$) and the scale factor ($414\,684$) is exactly the first-order T_CMB error – this confirms the correctness of the bridge formula.

Physical interpretation

ξ_{FFGFT} and ξ_{UIFT} are **not the same quantity**. They are connected by a pure SI conversion factor: the ratio of the relativistic electron energy ($m_e c^2 = 511$ keV) to the eV scale of the FFGFT temperature formula, modulated by $(16/9) \cdot \ln 2$.

The SI conversion table above was present in older versions of Doc. 013 (archived March 2026, Git commit `f2be9c8`) and has since been shortened. The explicit statement $T_{\text{CMB}}^{\text{nat}} = k_B \times 2{,}725$ K $= 2{,}35 \times 10^{-4}$ eV is found verbatim in that version. The reconstruction here restores the connection between the natural-unit formula and the SI value, which was implicitly assumed but no longer shown.

The Vopson-Teker experiment measures ξ_{UIFT} directly. If confirmed, that evidences the thermodynamic ground state at T_{CMB} – not $\xi_{\text{FFGFT}} = 4/30000$. The geometric derivation of ξ_{FFGFT} from $\kappa = 7$ (Doc. 009) is a separate and independent condition.

.17 Refinement 14a: Redshift in Doc. 041 – mechanism instead of energy loss

Affected document

Doc. 041 (parameter derivation), table “Cosmological Parameters”, LEVEL 3 (redshift) – in particular the description “photon energy loss through ξ-field”.

Previous formulation

The cosmological redshift is labelled in the comparison table as “photon energy loss through ξ-field” – i.e. as an energy loss of the photon in the ξ field.

Refined formulation

The physical mechanism is the **fractal path lengthening**: the geometric stretching of the propagation path in the ξ field (torsion lattice), as derived authoritatively in Doc. 026, Doc. 149 and Doc. 182. The dependence on the path length d (and the wavelength) remains and is consistent with the fractal path lengthening – it is *not* a contradiction.

An “energy loss” of the photon is **not an independent mechanism** but an *apparent artifact*: it would arise only if one erroneously fails to assume the geometric path lengthening. The wording “photon energy loss” in Doc. 041 is therefore to be replaced by “fractal path lengthening (geometric path stretching)”; the “energy loss” may at most be named explicitly as the artifact that would arise with the path lengthening omitted.

Justification

The static FFGFT cosmology explains the redshift purely geometrically (Doc. 182: “not by an intrinsic energy loss at the photon, but by geometric stretching of the propagation path”; *tired light* does not occur). The comparison table in Doc. 041 abbreviates this mechanism to “energy loss” and thereby evokes precisely the tired-light connotation that FFGFT does not intend. The refinement brings Doc. 041 into line with the authoritative derivation (Doc. 026/149/182) without changing the path-length dependence.

Status of incorporation

Incorporated (as of 2 June 2026). Implemented in Doc. 041 (DE+EN).

.18 Refinement 15: Mark cosmological z as a ΛCDM comparison quantity

Affected documents

Doc. 061 (temperature units/CMB), section “modified recombination history” as well as abstract/summary; Doc. 041 (comparison tables ΛCDM vs. T0).

Previous formulation

A z-dependent recombination calculation ($z_{\text{rec}}^{T0} \approx 1089{,}5$, $T(z) = T_0(1+z)[\dots]$) stood as "In T0 theory recombination occurs at ..." – i.e. as an FFGFT-internal quantity. Likewise "CMB at $z \approx 1100$" in the abstract/summary.

Refined formulation

The static ξ universe has *no* cosmological z in the expansion sense; FFGFT's redshift is fractal path lengthening, not a scale factor. Therefore z in these calculations is a Λ**CDM pipeline quantity** and is to be marked explicitly as a *comparison calculation*, not as an FFGFT result. In FFGFT the CMB arises through stationary thermalization in an infinitely old universe, not through decoupling at some z. In Doc. 061 the section is framed accordingly; the abstract/summary places are changed to "without decoupling at any z". In Doc. 041 the $z = 1100$ values are marked as "(ΛCDM)" (they stand in the ΛCDM comparison column anyway).

Justification

Unmarked comparison calculations create the impression of FFGFT-internal results and violate the standing caution that cosmological pipeline quantities (H_0, z, Λ_{obs}) must not be presented as FFGFT statements without labeling. The marking restores the separation without removing the comparison numbers.

Status of incorporation

Incorporated (as of 2 June 2026). Docs. 061 and 041 (DE+EN) marked. Further unmarked pipeline comparisons in the corpus are to be checked separately (ongoing).

.19 Refinement 16: H_0 is emergent (decompactified), not fundamental

Affected documents

Doc. 026 (geometric cosmology), l. 116/117 DE+EN; with offshoots in Doc. 064 (l. 104) and Doc. 181 (l. 158). *Already correctly implemented in Doc. 263* (H_0 as an emergent, decompactified quantity); the remaining places are here only *noted*, not yet changed.

Previous formulation (outdated)

Doc. 026 calls H_0 a "fundamental constant describing the interaction of light with the geometry of the T0 vacuum". This formulation stems from an earlier state that still understood the redshift as an *energy loss* of light passing through the vacuum (a near-"tired light" picture). Doc. 064 ("measures energy loss in the ξ-field") and Doc. 181 ("geometric energy loss") carry the same remnant.

Refined view

Two independent reasons argue against treating H_0 as a fundamental constant:

- **Energy loss is outdated (cf. R14a):** The redshift is fractal path lengthening, not energy loss. The "vacuum as a traversed medium" is itself already a decompactified description and not part of the compactified T^4 view.
- H_0 **is emergent, not fundamental:** Doc. 016 records that age, critical density and Hubble length are *Friedmann concepts without a direct T0 counterpart*; Doc. 262, that $\hbar, \varepsilon_0$ become *real only upon decompactification*. Since $H_0 = E_H/\hbar$ contains the (only then real) $\hbar$, H_0 does not exist in the compactified T^4 form. H_0 is an emergent quantity of the decompactified description; the non-integer exponent $41/4$ is an expression of this scale transition, not a fundamental geometry constant. (The integer ansatz $n = 10$ in Doc. 165 is therefore to be viewed with caution – it forces an emergent transition quantity into a compact geometry.)

Status of incorporation

Partially incorporated (as of 2 June 2026). Implemented in Doc. 263 (H_0 emergent/decompactified). **Open:** Doc. 026 (DE+EN, l. 116/117) is a document that is *outdated* in content (energy-loss/vacuum picture) and requires a more fundamental revision or a labeling as historical; likewise the offshoots in Doc. 064/181. Noted as a larger construction site.

.20 Refinement 17: circular/fit-dependent DE-DM relations in Doc. 028

Affected document

Doc. 028 ("7 questions"), sections "Riddle 4: MOND" and "Riddle 5: Dark Energy and Dark Matter". Only *noted*, not yet changed.

Finding (recomputed)

- **DE/DM ratio: circular as a *prediction*, admissible as a *conversion*.** Doc. 028 gives $\rho_{\mathrm{DE}}/\rho_{\mathrm{DM}} = \xi^\alpha$ with $\alpha = \ln(2{,}5)/\ln(\xi)$. Inserting this α into ξ^α yields *exactly* $2{,}5$ *for any* ξ (checked with $\xi = 10^{-3}, 10^{-5}, 0{,}5$). *As a prediction* from the ξ geometry the relation is thus circular (violates Doc. 101). *But:* the target value $2{,}5$ ($\Omega_\Lambda/\Omega_{\mathrm{DM}}$) is itself not a model-neutral measured value but a ΛCDM pipeline output (standing caution). Reading the relation therefore *not* as a prediction but as a *conversion factor* between FFGFT geometry and the ΛCDM-Ω bookkeeping – analogous to c (meter/second) and to z (cf. R15) – the circular calibration via $2{,}5$ is *admissible*. **Limitation:** unlike c (both sides have the same real counterpart), the ΛCDM side with ρ_{DE} contains a model artifact (Λ does not exist in FFGFT). The factor thus translates partly a real quantity (the DM bookkeeping $\leftrightarrow$ ξ effect), partly an artifact (ρ_{DE}). Therefore: admissible, but *only of limited* informativeness – as far as the ΛCDM quantity is artifact-free.
- **The MOND scale is fit-dependent.** $a_0/(cH_0) = \xi^{1/4} \cdot K_M$ gives the measured value $0{,}176$ only with $K_M = 1{,}637$. The $\xi^{1/4} = 0{,}1075$ is genuinely from ξ; K_M is a *free adjustment factor* without derivation. Partially derived, not parameter-free.
- **Two incompatible a_0 formulas; rotation curve open.** Besides Doc. 028, Doc. 201 (§4.3) gives a *different* form: $a_0 = c^2\xi/(4\lambda) \approx 1{,}2\times10^{-10}$ m/s^2. This is *not reproducible*: for the correct unit (m/s^2) λ would have to be a *length* ($\sim 2{,}5 \times 10^{22}$ m), whereas in the rest of Doc. 201 λ is via $m_T = \lambda/\xi$ an *energy/mass* – the same symbol in two incompatible meanings. Without an independent λ value, a_0 is either undetermined or (backward from the measured value)

circular; the length scale $2{,}5\times10^{22}$ m corresponds to no known FFGFT scale. *Consequence:* both a_0 formulas (028: fit-dependent; 201: dimensionally inconsistent) do not carry. The *flat rotation curve* follows qualitatively as a MOND consequence of the ξ geometry (static, without DM substance; Doc. 201/145/156), but the concrete profile $v(r)$ and the SPARC comparison are *not worked out* – remains open.

Delimitation (what remains valid)

The CMB relations are *not* affected and remain robust: $T_{\text{CMB}} = \frac{16}{9}\xi^2 E_\xi$ and $|\rho_{\text{Casimir}}|/\rho_{\text{CMB}} = \pi^2/(240\xi) \approx 308$ contain ξ alone, without a subsequently adjusted exponent or fit factor.

Status of incorporation

Noted (as of 2 June 2026), not yet changed. Recommendation: the DE/DM ratio (Riddle 5) is to be marked *not* as a prediction but as a limited conversion factor to the ΛCDM-Ω bookkeeping (admissible like c/z, but bounded by the ρ_{DE} artifact); the MOND scale (Riddle 4) as partially derived with a free K_M. The *interpretation* of the dark sector is settled – DM is a ξ-geometric effect (an artifact of the ΛCDM view, like H_0 and z), DE is eliminated (no Λ). *Quantitatively* it remains open: no model-neutrally derived Ω_{DM}, no worked-out rotation curve.

.21 Refinement 14b: CMB anharmonicity, forward vs. retrodiction, and the $|n|^2 = 30$ case (R29/R30/R31)

An explicit forward investigation (Doc. 268, step 17; scripts `Dok268_Skripte/`) clarified the status of the CMB peak sector and split it into three points.

(1) The direction of the inference (R29). The earlier presentation created the impression that $\{1,6,14,26\}$ and hence $\{3,18,42,78\}$ follow forward from the T^4 geometry. In fact $\{1,6,14,26\}$ is obtained by squaring the *measured* CMB ratios (Doc. 268, step 4). It is a **retrodiction** (consistency check), not a parameter-free prediction. The genuine forward maxima of the full T^4 spectrum are dense ($|n|^2 = 3,6,10,14,18,22,26,30,\ldots$).

(2) The $|n|^2 = 30$ case (R29, sharpened by O. Teker). $30 = 3\cdot10$ satisfies the factor-3 filter, since $|n|^2 = 10$ is a local maximum. Thermally one even has

$$I(30) = 13{,}2 \ > \ I(42) = 7{,}6 \ > \ I(78) = 1{,}7,$$

so 30 ($\ell \approx 696$) would be *stronger* than two visible peaks, yet shows no peak in the CMB (it lies in the trough between peak 2 and 3). The earlier argument "strongest local maximum in the range" fails here. The selection mechanism is thus *not* provided by the factor 3 alone. *Open.*

(3) The naive C_ℓ Bessel projection is insufficient (R30). The calculation $C_\ell = \sum_n N_{\text{BE}}\,|j_\ell(k_{\text{obs}}D_A)|^2$ does *not* reproduce $\{3,18,42,78\}$; it washes out to a quasi-continuum dominated by the lowest modes. The earlier R30 formulation ("a feasible calculation, no fundamental obstacle") is withdrawn: a physical source/window function is missing. *Open.*

(4) The forward content – geometric anharmonicity (new, R31). Forward, the symmetric T^4 resonance generates the *harmonic* backbone $1:2:3:4:5$ ($k_{\text{obs}}^2 = 3n^2$). The CMB is *anharmonic* ($1:2{,}44:3{,}68:5{,}09$). This anharmonicity follows forward and parameter-free from the codimension-1 projection:

$$D_{\text{eff}} = D_f - 1 = 2 - \xi \approx 2 \ \Rightarrow\ \nu = \frac{D_{\text{eff}}}{2} - 1 = -\frac{\xi}{2} \approx 0 \ \Rightarrow\ J_0 \text{ membrane} \ \Rightarrow\ 1 : 2{,}30 : 3{,}60 : 4{,}90,$$

matching to $\sim 5\%$, *without* mode selection. The radial Bessel index is $\nu = D/2 - 1$; the anharmonicity is maximal at $D = 2$ and vanishes at $D = 1$ and $D = 3$. The CMB sits at the dimension of maximal anharmonicity. The match is to the *flat* membrane (J_0), not to the curved 2-sphere ($Y_{\ell m}$, gives $1 : 1{,}73 : \ldots$, wrong). This is the conceptually clean, forward-directed statement; it removes the circularity objection because no selection from data is needed.

(5) The residual deviation – not from ξ. The best fit lies at $D \approx 1{,}86$ (just below 2). A spectral dimension $d_s < D_f$ would be the obvious source but fails: the FFGFT fractal is too weakly fractal ($d_w = 2 + O(\xi) \Rightarrow d_s \approx 2$); for $d_s = 1{,}86$ one would need $d_w \approx 2{,}15$ ($\sim 1000 \times \xi$). The deviation is not derivable from ξ and is of the order of the measurement uncertainty (J_0 is consistent with peaks 3,4 within $\sim 2\sigma$; only peak 2 at $\sim 4\sigma$). Pointing to an exact value would be number-juggling. *Open, presumably partly measurement noise.*

Consequence. The defensible core reads: harmonic T^4 backbone (forward) + geometric anharmonicity from the codimension-1 projection (forward, $\sim 5\%$). The ratio match $\{3, 18, 42, 78\}$ is a retrodiction; the rigorous single-peak selector and the exclusion of $|n|^2 = 30$ remain open. The particle sector (α, lepton masses, Koide) is *unaffected* by this – it rests on direct ξ powers, not on mode counting.

No.	Doc.	Erroneous / unclear	Correct / refined
C1	049 (HTML)	$\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2)$	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$
C2	116 (tab.)	$r_\tau = 8/3$	$r_\tau = 25/9$
C3	018 (Explorer)	$a_e = 4\pi/(f \cdot k)$	$a_e = 2\pi^2/(f \cdot k)$
C4	267 (DE+EN), 268 (DE)	factor 3 interpreted as the fractal dimension D_f ("fractal dispersion relation $k_{\text{eff}}^2 = D_f k^2$", Doc. 051)	**Factor 3 = three observable space dimensions:** the observer measures $k_{\text{obs}}^2 = (n_1^2 + n_2^2 + n_3^2)/L^2$ (the fourth torus direction is time unfolding). The conformal metric (Doc. 051) cancels for massless particles; the fractal correction (Doc. 204) is $+0{,}69\,\xi k^2 \approx 10^{-5}$, *not* the factor 3. $D_f = 3 - \xi$ is consistent with factor 3 but is not its origin. *Done (267 DE+EN, 268 DE), June 2026*
C5	275 (script v3)	Three implementation errors in `ffgft_hlv_gap_v3.py`: (a) the T^4 point lattice used n_3 only in the acceptance norm, not in the position (729 unique out of 2000 points; the $1/d^2$ clamp produced $\lambda_{\text{max}} = 2{,}5 \cdot 10^{12}$) – the result "gap_CV $= 1{,}68$ outside the band" was an artifact; (b) `hlv_points(seed)` did not use the seed – the 3-seed "ensemble" was the same calculation three times; (c) the shell detector for target ratios < 1 was structurally biased (minimum error $(1{,}05/c - 1)$; a larger sub-1 ratio wins on every point set) – the discrimination $\Delta = 0{,}145$ was structural	**All three fixed in v4** (11 June 2026): T^4 as kNN in genuine 4D coordinates with dimension-matched 4D nulls (result: $1{,}329 \pm 0{,}025$ *inside* the band $[0{,}690, 2{,}406]$); HLV ensemble with $\sigma = 0{,}05$ ($0{,}710 \pm 0{,}008$, in the band); the shell detector raises an exception for targets ≤ 1. v3 results in Doc. 275 visibly retracted. *Audit: P. Stoychev (IPI), 11 June 2026*
R1	116 / 158	$\Delta Q < 0{,}00003\,\%$ vs. $0{,}001\,\%$	both correct; $0{,}001\,\%$ for external use
R2	several	$\hbar$ as a postulate	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
R3	173–176	a single uniform ξ value	two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)
R4	180–182	L_0 as a Schwarzschild radius	L_0 = geom. fundamental length
R5	several	"FFGFT computes masses exactly"	mass structure from ξ; $\sim 1\,\%$ agreement
R6	116 / 158	Koide with FFGFT symbolic expressions	Koide only with numerical values (non-closure, Doc. 193)
R7	005, 008, 019, 086, 189, 202	"renormalization group" as an FFGFT tool	scale structure from the recursion $\mathscr{R}$
R8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	"fractal renormalization" as an internal term	geometric scale adjustment / fractal correction

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No.	Doc.	Erroneous / unclear	Correct / refined
R9	005, 006, 042, 046	column “symmetry” with “color”; $N_c = 3$ as an external SM input	columns “scaling class” + “SM correspondence”; quarks via $m = r\xi^p v$
R10	009	$K = 245$ from $\varphi^{12}/1{,}314$; “no free parameters”	$K = R_{pe}/(4/3)^7$: an empirical anchor; factor $1{,}314$ defined implicitly, not derived
R11	025, 003, 133	K_{frac} closes the T_{CMB} gap; one definition of K_{frac}	correct: $(1 - 275/4 \cdot \xi)$, $\Delta = 0{,}0002\,\%$; Doc. 003 and Doc. 133 use different models
R12	022, 230	ΔCHSH $\approx 10^{-5}$ (Doc. 230) vs. 5×10^{-4} at $N = 73$ (Doc. 022)	two different observables: cumulative (Doc. 022) vs. elementary $\xi/(2\pi)$ (Doc. 230); 10^{-5} unreachable at any finite N from Doc. 022
R13	013, 025, 061	$\xi_{\text{FFGFT}} = \xi_{\text{UIFT}}$; SI conversion table missing	$\xi_{\text{FFGFT}}/\xi_{\text{UIFT}} = m_e c^2/(\frac{16}{9}\ln 2 \cdot \text{eV}) \approx 414\,684$; table in Doc. 013 (archived) reconstructed
R14a	041	redshift as “photon energy loss through ξ-field”	fractal path lengthening (geometric path stretching); energy loss only an artifact when the path lengthening is omitted; path-length dependence remains (Doc. 026/149/182); the same path lengthening carries the CMB treatment in Doc. 268 (formerly tracked separately as R19)
R15	061 / 041	cosmological z ($z \approx 1100$, $z_{\text{rec}} \approx 1089{,}5$) unmarked as an FFGFT quantity	marked as a ΛCDM comparison quantity; the static universe has no cosmological z; CMB by stationary thermalization
R16	026 / 064 / 181	H_0 as a “fundamental constant” (energy-loss/vacuum picture)	H_0 is emergent (decompactified), not fundamental; implemented in Doc. 263, Docs. 026/064/181 noted as outdated
R17	028	DE/DM ratio $\xi^{\ln(2{,}5)/\ln\xi}$ as a prediction; MOND with fit factor K_M	the DE/DM relation is circular (gives 2.5 for any ξ); MOND partially derived ($\xi^{1/4}$ genuine, K_M free); CMB relations unaffected; noted
R18	267	the relation ΛCDM expansion vs. FFGFT time unfolding treated as an empirical question	the two pictures are *not* distinguishable through the observed ratios; the decision is theoretical, not empirical (conceptual placement)
R20	267 / 268	absolute cosmological scale (R_H, D_A, z_*) presented as an FFGFT result	the *form* is fixed: a dimensionless ξ ratio (R_H/ℓ_P). The *factor* (exponent $41/4$, $E_H = E_0\,\xi^{41/4}$, Doc. 182) is *not* derived from the T^4 geometry and must not be fitted (fitting = retrodiction). FFGFT is not the traditional yardstick; the factor is therefore adopted as a *declared external calibration* from ΛCDM – a choice of units, not an adoption of physics – and inserted with the converged value once the H_0 tension (~67 vs. ~73) is resolved; parametrized until then. The imported value does *not* count as FFGFT confirmation. Affects only absolute positions ($\ell_1 \approx 220$, $z_* \approx 875$), not the ratios; the internal forward derivation of the factor remains the goal. *Open*
R29	268	selection rule: why does the CMB show only the subset $\{3, 18, 42, 78\}$ of the T^4 peaks?	*Only partially answered.* The factor 3 from the 3D projection ($3 \cdot \{1, 6, 14, 26\}$) is plausible, but the sequence $\{1, 6, 14, 26\}$ is *read off the data* (Doc. 268, step 4), not produced forward. In particular $\lvert n\rvert^2 = 30 = 3\cdot 10$ is *not* excluded: $\lvert n\rvert^2 = 10$ is a local maximum, and thermally $I(30) = 13{,}2 > I(42) = 7{,}6 > I(78) = 1{,}7$ – so 30 ($\ell \approx 696$) should be a peak, yet sits in a CMB trough. The “strongest in range” argument fails. *Open*, sharpened by O. Teker

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No.	Doc.	Erroneous / unclear	Correct / refined
R30	268	rigorous justification of the selection mechanism (full C_ℓ projection) missing	the *naive* spherical Bessel sum $C_\ell = \sum_n N_{\text{BE}} \lvert j_\ell(k_{\text{obs}} D_A)\rvert^2$ does *not* reproduce $\{3, 18, 42, 78\}$ – it washes out to a quasi-continuum dominated by low modes. The earlier formulation "a feasible calculation, no fundamental obstacle" was too optimistic: a physical source/window function selecting the peaks is missing. *Open*
R31	268	anharmonicity of the CMB peaks: whence the stretch relative to the harmonic $1:2:3:4:5$?	*Leading behaviour derived forward.* Forward, T^4 gives the harmonic backbone $1:2:3:4:5$ (symmetric resonance $k^2 = 3n^2$). The observed anharmonicity $1 : 2{,}44 : 3{,}68 : 5{,}09$ follows from the codimension-1 projection: last-scattering surface $\Rightarrow D_{\text{eff}} = D_f - 1 = 2 - \xi \approx 2 \Rightarrow$ Bessel index $\nu = -\xi/2 \approx 0 \Rightarrow J_0$ membrane $\Rightarrow 1 : 2{,}30 : 3{,}60 : 4{,}90$, matching to $\sim 5\%$, parameter-free, without mode selection. The small residual ($D \approx 1{,}86$) is *not* derivable from ξ (the spectral dimension of the weakly fractal space stays ≈ 2) and is of the order of the measurement uncertainty. *Partially derived*; remainder open
R32	corpus-wide	statements presenting H_0 or the absolute scale as *derived by FFGFT*	follows from R20: H_0 is an *imported* external calibration, not derived. All passages presenting H_0 as an FFGFT result are to be cleaned up ("imported/declared" instead of "derived"). *Noted for now*; corpus-wide cleanup deferred. *Open*
R33	009 / 042 / 181	magnitude factor 5^4 in $\xi = 1/(3 \cdot 2^2 \cdot 5^4)$ presented as *derived*	the *form* of ξ is forward: the harmonic $4/3$ (perfect fourth, Doc. 042/159) gives the leading digit, 3 (space dimensions) and $2^2{=}4$ (spacetime) are read off the geometry. The *magnitude-carrying* factor $5^4{=}625$ ($\rightarrow 10^{-4}$), however, is *not* derived forward: it is merely *asserted* ("emerges from the fractal structure", Doc. 009; "encodes the symmetry structure", Doc. 181 – there circular via $30000/4$) or the value is *empirical* (Doc. 042: measured from the Higgs sector $1{,}3194{\times}10^{-4}$, ~1 % off the ideal $4/30000$). It remains the *intuitive* (harmonic / Euler Tonnetz) resp. empirical grounding of the magnitude; this is openly *declared* as such, not presented as a forward derivation. Structural parallel to R20 (form fixed, factor open), one level deeper. *Declared/noted*

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No.	Doc.	Erroneous / unclear	Correct / refined
R34	181 / 013	$D = 4$ presented as "following necessarily / not a postulate"; role of the SI conversions for the c-/time-power	*3+1 is empirical input* – no theory derives the dimension count; the earlier "follows necessarily" claim (Doc. 181) is overstated. Forward are (i) periodicity (divergence-freedom), (ii) torus-not-sphere ($\pi_1 \neq 0$, stable windings), (iii) *minimal sufficiency* – 4 instead of a naive 5, because the duality $\tilde{T}{\cdot}m = 1$ lets one circle carry mass *and* time, (iv) the lossless (bijective) Hilbert representation (Type III, Doc. 270/206/230). The *bare number* 4 is *not* forced forward (the "4 from ξ" argument in Doc. 181 is circular resp. base-dependent). *Connection with the units:* the four-circle structure is at the same time the exponent bookkeeping of the SI conversions – c = space-time bridge (the three T^3 circles against the time direction), $\hbar$ = mass-time bridge (S^1_m, $\tilde{T}m = 1$). Hence the SI factors are indispensable for the *power of* c and the *power of time*: $c = \hbar = 1$ collapses L, T, M to one axis and destroys exactly the exponent structure one wants to read off. T^4 (mass reading) and T^3+time (time reading) are the same object. Signature as R20/R33: structure/form forward, number input. *Clarified*
R35	corpus-wide (esp. 182, 250s, scale tables)	quantities of the form $X = \xi^N$ resp. $X = \varphi^N$ read as FFGFT confirmation; the SI conversion factor (ℓ_P) partly tacitly absorbed into the ξ exponent	ξ^N **triviality:** "$X = \xi^N$" is by itself *vacuous*, since any magnitude can be written as a ξ power – an exponent alone is not a prediction (cf. R10, R20, R33: form forward, number input). The SI conversion factor ℓ_P must *not* be absorbed into the exponent. **Finding:** only $L_0/\ell_P = \xi^1$ is exact; $R_H = \ell_P \cdot \xi^{-63/4}$ with $63/4 = 41/4 + 11/2$ – the corpus $41/4$ is referenced to E_0 instead of the Planck scale; the "nice fractions" are roundings of measured exponents (measured $\approx 15{,}72$ vs. $63/4 = 15{,}75$). A ξ^N representation counts as content only if N is derived forward. *Addendum (Doc. 279, 14 June):* the $11/2$ in $63/4 = 41/4 + 11/2$ is *not arbitrary* – it coincides with the forward-derived fine-structure exponent $\alpha = \xi E_0^2 = K\xi^{11/2}$ (Doc. 011/070, the same $E_0 = \sqrt{m_e m_\mu}$ as the $41/4$ anchor), to $\sim 0{,}5\,\%$ ($E_0/E_{\text{Planck}} = \xi^{5{,}48}$ vs. $5{,}5$). *A conjecture, not evidence*: different reference bases (MeV vs. Planck), the identity proven only by $E_0 = E_{\text{Planck}}\,\xi^{11/2}$ forward. The cosmic $41/4$ remains open. *Declared (changelog Update 16), table entry added 10 June 2026*
R36	182 / cosmol. sector	R_H interpreted as the "maximal scale" resp. size of the universe	R_H **interpretation refined:** $R_H = c/H_0$ is the *Hubble length* – the local expansion rate expressed as a distance – *not* the size of the universe and *not* the observable horizon (the latter $\approx 3\,R_H$ in ΛCDM). The actual universe exceeds R_H; only a *lower bound* is measurable. The "maximal scale" from Doc. 182 is tenable only as the upper scale of the *cosmological torus sector*, not as the extent of the universe. Consistent with R20 (R_H calibration as a declared external input). *Declared (changelog Update 17), table entry added 10 June 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
R37	009, 188, 271, 272, 275	φ in various documents in various roles: (a) phenomenological scaling factor between fractal levels (Doc. 188); (b) substrate geometry of HLV, not FFGFT (Doc. 271); (c) trivial representation $X = \varphi^N$ as a non-prediction (Doc. 272); (d) $\varphi^{12}/(1{,}314)$ with an empirical anchor (Doc. 009, R10)	**Three levels clearly separated:** **(1) Microscopic – input:** $\xi = 4/30000$ (fractal dimension). φ is *not* active here; ξ and φ are different objects. **(2) Macroscopic – revised (see R38):** $1/\varphi \approx 0{,}6180$ is a *pass-through point* of the ξ damping recursion $r(k+1) = r(k) \cdot (1-\xi)$ at $k^* = \log(\varphi)/\xi \approx 3609$ (the only fixed point: 0). The original derivation claim is downgraded to an open question – $k^*(c)$ exists for any target c, R35 applies (details: R38). The scale interpretation stands: levels with $q = 2/3$ internally convergent, no RG steps with factor 2. **(3) Particle physics – output:** $Q_{\text{FFGFT}} \approx 0{,}6677$ (Koide scalar, Doc. 258) operates only on the lepton-mass scale. Not to be confused with $1/\varphi$. **Warning Doc. 272 / R35 / R10:** the warning "any quantity is writable as φ^N – not a prediction" remains valid – and, after the revision of 11 June 2026, also *applies to* the k^* result from Doc. 275 (the earlier opposing delimitation here is retracted, see R38). *Declared; level (2) revised 11 June 2026*
R38	275	$k^* = \log(\varphi)/\xi \approx 3609$ presented as a "fixed point" result "without a free parameter"; delimitation from the φ^N triviality (R35) claimed with three arguments	**Status clarification:** (a) $1/\varphi$ is a *pass-through point*, not a fixed point – the only fixed point of $r \mapsto r(1-\xi)$ is 0. (b) $k^*(c) = [\log c - \log(D_f/3)]/\log(1-\xi)$ exists for *any* target $c \in (0,1)$ with identical parameter status: $k^*(Q_{\text{FFGFT}}) = 3029$, $k^*(1/2) = 5198$, $k^*(1/e) = 7499$, $k^*(\alpha_{\text{em}}) = 36899$ – the free exponent was merely renamed k^* and solved for the target. R35 applies in full (first corpus-internal self-application of the guardrail). (c) The arithmetic ($r(3609) = 0{,}6179940$, error $4 \cdot 10^{-5}$) is correct; the observation would gain physical content only through an independent fixing of $k^* \approx 3609$ (level counting, observable) – currently open. *Revised after audit P. Stoychev, 11 June 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
R39	182, 267, 277, corpus-wide (H_0/R_H)	H_0/R_H calibration (R20/R36) readable as an FFGFT-specific gap resp. as an asymmetric disadvantage relative to ΛCDM	**H_0/R_H status refined (shared reference point, not an FFGFT gap):** the relation Planck length↔ R_H (exponent $41/4$, $R_H/L_0 = 6{,}49 \times 10^{64}$) is derived from first principles by *no* framework – why the universe is $\sim 10^{60}$ Planck lengths across is predicted by no theory. ΛCDM fixes the relation via the *measured* H_0 – itself circular in the sense of the data degeneracy (Doc. 267). FFGFT adopts the established value for comparability and writes it in ξ notation. Hence the cosmic scale is a *shared empirical reference point of both models*, *not* a free parameter and *not* an FFGFT-specific gap. Reinforcing: in FFGFT $\hbar, c$ are SI conversion and G is derived ($\xi \to \{m_e, ...\} \to G \to \ell_P$); a "measured" dimensional anchor in the standard-physics sense does not even exist here. The earlier picture "imported/to be cleaned up" (R20/R32) remains correct but is *not* to be read as a weakness relative to ΛCDM. **Remaining open point (symmetric):** solely the *forward derivation* of the exponent $41/4$ from ξ (R20). It is missing – but it is missing for every framework, not only FFGFT. The only genuine model asymmetry is the *dark sector* ($\Omega_\Lambda, \Omega_{\text{dm}}$ as free densities), carried by ΛCDM alone (Doc. 277, cost table). *Declared 14 June 2026*
C6	004, 025, 030, 039, 041, 053, 061, 063, 068, 081	Cosmological redshift presented as *wavelength-dependent* ($z(\lambda) \propto \lambda$, "UV stronger than radio"; e.g. Doc. 063 $z(\lambda_0) = \frac{\xi x}{E_\xi}\lambda_0$; Doc. 061 $\Delta z = 0{,}008\ln(\lambda/\lambda_0)$; Doc. 068 $z(\lambda) = z_0(1 + \ln(\lambda/\lambda_0))$).	**Redshift is achromatic.** The fractal path-lengthening is purely geometric and stretches *all* wavelengths by the same factor: $1 + z = e^{\xi x}$, independent of λ. A *finite-element simulation* of the geometry yields no wavelength dependence; the leading term $z = \xi x$ is frequency-independent (Doc. 064). The observed cosmological redshift is achromatic – FFGFT gets this right, and the earlier chromatic "discriminator" drops out (it moves to the "fits" column, data degeneracy Doc. 267). Worked out in Doc. 280. *Declared 15 June 2026*
R40	005, 006, 011, 012, 133, 275	Implicit: from ξ the *absolute values* (masses, α, G) follow directly and exactly; the *100* in K_{frak} and the *3609* of the φ-recursion are taken as related/fundamental.	**Only ratios are exact.** Absolute values carry the fractal/SI correction, absorbed into the bridge constants v (masses $m = r\xi^p v$), $E_0 = 7{,}398$ MeV (α) and $3{,}521 \times 10^{-2}/2{,}843 \times 10^{-5}$ (G; `calc_De.py`); in ratios they cancel ($m_\mu/m_e = 206{,}768$). Factored out $= K_{\text{frak}} = 1 - 100\xi$ (Doc. 011/012; Doc. 133: from D_f-consistency + RG over 17 orders, "not an adjusted correction"). *Keep separate:* **100** = RG amplification (Doc. 133); **3609** = depth at which $r(k) = \frac{D_f}{3}(1 - \xi)^k$ passes the *passage point* $1/\varphi$ (Doc. 275) – not a fixed point, unanchored unless $1/\varphi$ is independently motivated (open, R37). Further couplings (backdoor) open; clean falsification only via forward prediction ($41/4 \to H_0$). *Declared 15 June 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
P41	267, 280, 221 (cosmological sector)	Redshift drift (Sandage–Loeb) proposed as a possibly sharp discriminator of static ($1+z=e^{\xi x}$, $\dot z=0$) versus expansion – a *new* observable rather than a re-reading of the same data, potentially piercing the degeneracy (Doc. 267).	**Checked and not booked (degeneracy risk at the shared scale).** ΛCDM signal $\Delta v = c\,\dot z\,\Delta t/(1+z)$ over 25 years: $+6$ cm/s (z=0.5), sign change at $z_0=1.92$, -14 cm/s (z=4), -20 cm/s (z=5); ELT/ANDES order $\sigma\sim 2$ cm/s. The *amplitude* test is not degeneracy-proof: a secular evolution of the phase ξx at rate $s\approx(0.1–0.4)\,H_0$ mimics ΛCDM, and the natural FFGFT rate is $c/R_H=H_0$ *exactly* – both models share the scale (P36/P39); moreover the dipole systematic of the Galactic acceleration (≈ 17 cm/s over 25 yr, direction-dependent and subtractable) lies at the same order as the signal. Conditionally sharp remains only the *shape* test: the sign change at z_0, i.e. the two-band contrast $\Delta v(0.5)-\Delta v(4)\approx+20$ cm/s, cannot be mimicked by *any* z-constant secular channel – but it requires a corpus-level *advance* commitment to strict staticity ($\dot z=0$) and low-z carriers beyond Lyman-α; time horizon ~2 decades. No falsification criterion booked; calculation: `Werkzeuge_Skripte/drift_sandage_loeb_check.py`. *Declared 2 July 2026*
P42	282, 292 (lepton sector)	Doc. 282 presented the pair $(\sqrt{2},\ 2/9)$ as *two pure numbers without a fit*; the precision analysis (Doc. 292) shows that the circulant with exactly this pair misses m_μ/m_e by $1.0\cdot10^{-5}$ at a measurement precision of $2.2\cdot10^{-8}$ – excluded as an exact pair at the pole-mass level ($\sim 450\,\sigma$ in the μ/e-precise direction).	**One declared reference point: the ratio m_μ/m_e.** The framework fundamentally keeps the purely theoretical derivation from ξ; the lepton sector receives – analogous to R39 (H_0/R_H as shared empirical reference) – exactly *one* declared reference point: m_μ/m_e fixes, at structural $r=\sqrt{2}$ (from $Q=2/3$), the operative angle $\theta_{\text{ref}}=0.2222220471$. The structural number $2/9$ is its *rational address* to seven digits ($\lvert\theta_{\text{ref}}-2/9\rvert=1.75\cdot10^{-7}$), not an exactness claim at the pole-mass level. The $450\,\sigma$ finding thus dissolves in the claim layer. Two delimitations belong to this: *first*, the status of ξ is untouched – FFGFT claims ξ as forced by the geometry; that dispute is not adjudicated here. *Second*, the lepton masses are in themselves *not an input of the theory*: the structural numbers ($\sqrt{2}$ from $Q=2/3$, $2/9$ as address) stand on the theory side; the reference point belongs to the *comparison level* – it locates the geometric relation in the pole-mass/SI data (analogous to R39/R40: shared reference point resp. bridge constant) and would not be needed at all without the intent to compare. The honest price therefore lies not in the parameter count of the theory but in the *test balance*: m_μ/m_e is consumed as anchor and drops out as a test quantity. The gain: the m_τ prediction becomes input-error-free sharp, $m_\tau = 1776.9690\pm0.0001$ MeV ($+0.91\,\sigma$ against PDG 1776.86 ± 0.12; the Belle-II world average decides). For the bridge work (Doc. 291) the question sharpens accordingly: the dynamics must select θ_{ref}; $2/9$ remains its address and the pre-registered Stage-10 target (orbit resolution $\sim 0.1 \gg 10^{-7}$, unaffected). *Declared 2 July 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
P43	006, 011, 292 (`calc_De.py`)	Open question: whether the K_{frak}-with-100 formulas in the calculator `calc_De.py` need reworking after the generation-linear finding (Doc. 292, $N_g = g \cdot N_0$).	**No rework; recorded as a note.** Verified: `calc_De.py` does *not* use $K_{\text{frak}} = 1 - 100\xi$ explicitly. *(i)* $\alpha = \xi E_0^2$ with $E_0 = 7.398$ MeV (= the geometric route-2 value, Doc. 292); since $7.398/7.348 = K_{\text{frak}}^{-1/2}$, K_{frak} is already *absorbed* into the E_0 choice ($1/\alpha = 137.035$, no separate factor) – so the computation is already consistent with the two-route picture (Doc. 292), no contradiction. *(ii)* The masses run through per-particle (r, p) ($m = r\xi^p v$; e/μ/τ errors $1.18/0.66/0.37\%$) – *that* is the coarse ξ ladder in which the generation-linear law $N_g = g \cdot N_0$ (Doc. 292) applies. Replacing the (r, p) by N_g would be a gain only after $N_0 \approx 38.6$ is derived (otherwise fit against fit, P35); until then the (r, p) stand. No falsification/correction entry; the **100** in K_{frak} (fixed RG scale, P40) and N_0 (per-generation ladder factor) stay separate. *Declared 3 July 2026*
R41	284 (cf. 270, 249)	Doc. 284 (and the private Medin note) framed "every conversion to SI is itself a measurement" and "a geometry-specific memory cannot appear in any SI-carrying result" – too strong.	**The loss is not the SI conversion.** Per Doc. 270 the Type I duality decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T} \cdot m = 1$) is a *mode-preserving embedding* (lossless); compactness survives as the discrete spectrum and as physical observables (T_{rev}, CHSH$\sim \xi$). The genuinely lossy steps are the *reverse* compactification $\mathbb{R} \to S^1$, the spatial projections $D3 \to D0$, and – for a read-out – the *finite, non-periodic window plus decoherence* (Test G). FFGFT *does* carry framework-specific SI observables: mass reading (masses/Koide/$g-2$), CHSH$\sim \xi$, D_f, L_0 (Doc. 270/249). *Narrowed claim:* the coherence-sector recurrence (back-flow $5{,}125$, Doc. 282) is not *resolvable* on a warm, finite-window SI-time observable such as EEG – not "absent from any SI result". Doc. 284 (DE+EN) aligned accordingly. *Declared 19 June 2026*
R42	270 (cf. 248)	The selection of which T^4 circle becomes time was left implicitly open / treated as a bare empirical input.	**Not an open geometric question – it is the measurement.** In the compact view the four circles are symmetric (no space/-time distinction); time is not selected by the geometry but is *relational* – the symmetry-breaking *is* the measurement (the Type I decompactification = read-out, Doc. 270/284). The embedded observer experiences one direction as duration (Doc. 248, self-referentiality). Bookkeeping closes with no gap: one-ness of time = signature/predictivity (hyperbolic vs. ultrahyperbolic); *which* circle = the measurement / experienced situation; $3+1$ = the observer's situation, not a free parameter. New section added to Doc. 270 (DE+EN). *Declared 19 June 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
R43	285, 283	Doc. 285 first stated the 3-vs-6 relation as "inequivalent structure / structural fork, not a coordinate artifact."	**Corrected to containment.** $C_3 < A_5$ (3-fold and 5-fold both permute the same icosahedron, `recompute_reconcile.py`) and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$), so HLV $\supset$ FFGFT at the group/dimension level – the models are reconcilable, not incompatible. The $r(\tau) \leq 1$ result shows only that one cut-and-project map does not interconvert $\sqrt{2}$ and $2/\sqrt{5}$. The genuine difference is the *realised protected geometry*: FFGFT protects an independent 3-fold ($\sqrt{2}$, Koide $2/3$, confirmed); HLV the 5-fold ($2/\sqrt{5}$), and (Doc. 283) $\sqrt{2}$ is not among HLV's protected directions. Containment makes them compatible without identical. *Declared 19 June 2026*
R44	272, 190 (P35), 285	Previously: φ had no distinguished role (P35: any quantity is writable as ξ^N *or* φ^N – a cautionary example with no predictive content).	**Sharpened (no contradiction).** φ acquires meaning *not* inside FFGFT but as the derived eigenvalue of HLV's 5-fold / perp sector (golden inflation $\varphi^2 = \varphi + 1$) – a marker of the boundary HLV$\setminus$FFGFT. The same crystallographic restriction ($2\cos(2\pi/5) = 1/\varphi \notin \mathbb{Z}$) that forbids φ in FFGFT assigns it to HLV. P35 stands: φ as a fitting base carries no content, φ as a symmetry eigenvalue is structure, not a measured scale in an exponent – φ does not become an FFGFT prediction. Dot Theory classification (Doc. 272): φ = HLV-side discriminator of the structural nesting, entered as a contestable $O2$ rendering. *Declared 19 June 2026*
R45	285, 282–284	Doc. 285 stated the d7 spectrum point too narrowly (read like "quasicrystal spectrum without information content").	**Sharpened.** A quasicrystal has two spectra: the *structure / diffraction* spectrum is pure point (sharp Bragg peaks, $\approx 21\times$ above a random null, `quasicrystal_information.py`) – that is the information and exactly HLV's substrate-distinguishability (stands); the *energy / Laplacian* spectrum is singular-continuous (no sharp levels) – hence no masses. P35 caveat is narrow: only fitting *one* target number as φ^N is content-free. Honest residual: null-reproducible was the *temporal* over-damped kernel (Doc. 282–284), not the spatial structure. *Declared 19 June 2026*

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No.	Doc.	Erroneous / unclear	Correct / refined
R46	296 (cf. 025, 026, 063, 156)	Doc. 296 §5 ("Speculation: unobservable intermediate ranges") books dark matter/energy as *possible signatures of unobservable intermediate-scale ranges* – formulated as open speculation.	**Open check-point (not yet processed).** This intermediate-scale formulation potentially runs *counter* to the already established corpus line, which treats dark matter/energy not as an open signature but as a *derived consequence*: Doc. 025 reads dark matter as a ξ-field effect and dark energy as "not required"; Doc. 026/063 hold that the universe does not expand and dark energy becomes superfluous (H_0 reinterpreted); Doc. 156 frames the entire observable reality as the projection of a single energy field with static redshift $z \approx \xi \ln(d/\ell_P)$. The *intermediate-scale* mechanism is a different explanatory route from the established ξ-field route and is not a load-bearing concern of the framework. To be reconciled in a later revision of Doc. 296: either align the intermediate-scale formulation with the ξ-field derivation or mark it explicitly as a subordinate additional consideration. It should also be noted that the observer/projection/emergence core of Doc. 296 is already established in the corpus (Doc. 156) and is not newly introduced there. No correction entry, no rebuild – booked as an open note. *Declared 6 July 2026*
R47	285, 297, 298 (cf. 286, 284, 295)	R43 booked "HLV ⊃ FFGFT dimensional" ($D7 = D4 + 3'$-window). Read in the *time sector*, this pure dimension count treated Marcel's spiral-time window as *additional* – as if it sat *beside* FFGFT's recursion time and enlarged HLV.	**In the time sector: containment, not enlargement.** In unified notation – both $D6{+}$time, Marcel as $D6{+}$spiral-time, FFGFT in Hilbert space as $D6{+}$recursion-time – the spiral-time window is not additive but *internal*: a bounded section (window) of the unrolled recursion time. Hence HLV ⊂ FFGFT *in the time sector*. Precisely, Marcel's object is the over-damped $-1/\varphi$ branch (the $3'$-window, Doc. 285/286: carries structure, no mass), *not* the full recursion operator R – R has two branches, the relevant one (φ, $D4$, masses, revival 5.125) is FFGFT's. The embedding ι (Doc. 297, open conjecture) is exactly the map realizing this window as a bounded projection of the compact recursion. *Distinction from R43 (no contradiction):* R43's ⊃ is the pure *dimension count* (the $3'$-window as additional); R47's ⊂ is the *derivation content in the time sector* (the window as a section of the named compact whole – FFGFT supplies the whole, HLV evaluates the section without naming it). Both readings coexist *as long as both are read in the same $D6{+}$time notation*. *OP8-consistent:* a warm, finite EEG window over a partial turn looks locally Archimedean (Doc. 295) and does not resolve the winding (R41, Doc. 284) – that U_3 adds nothing resolvable over U_2 is thus the *prediction* of the window embedding, not a failure. *Open (symmetric):* whether ι is *forced* (genuine HLV ⊂ FFGFT in the time sector) or merely *allowed* (compatibility) – forcing test like C3A5-BRIDGE2, on the time axis; ι remains a conjecture (Doc. 297). *Declared 8 July 2026*